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Department of Computer Science & Engineering



Project Report on **“Online Bidding System”**

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1 INTRODUCTION

1.1 OVERVIEW OF THE SYSTEM

The BidMaster Platform is a web-based platform designed to facilitate real-time auctions in a secure and user-friendly environment. The system allows admins to create and manage auctions, while registered users can participate by placing competitive bids on listed items. This is built using the Laravel framework, the platform ensures reliable authentication, access control and smooth interaction between bidders and auction data. Each auction across five categories namely gadgets, artworks, antiques and collectibles, memorabilia and automobiles remain active for a defined duration, during which users can submit bids. The current highest bid is updated dynamically to maintain a transparent and engaging bidding process. Once an auction closes, the winning bidder is automatically notified via email.

This system eliminates the limitations of traditional physical auctions by providing accessibility, automation and efficiency, making the bidding process more convenient for both administrators and participants.

1.2 SIGNIFICANCE AND MOTIVATION

The motivation behind developing BidMaster is from the growing demand for efficient, accessible and transparent auction platforms. Traditional physical auctions are often limited by location, time and human coordination, making them impractical for many participants. Even existing online platforms may lack real-time responsiveness, simplicity, or affordability, especially for small-scale sellers and organizations. By using modern technology, the platform provides a flexible, user-friendly system that combines browsing and listing management with structured actions.

2 PROBLEM STATEMENT

Traditional bidding processes often lack transparency, speed and accessibility, leading to inefficiencies and reduced trust among participants. Existing systems rarely support real-time interaction or diverse auction formats in a user-centric manner. There is a need for a modern, reliable, and fair bidding platform that enhances engagement, ensures integrity and simplifies auction management.

2.1 ADDRESSING KEY SYSTEM GAPS

The BidMaster project addresses several critical gaps present in existing online bidding and auction systems. Many current platforms either lack real-time responsiveness, are overly complex for general users, or impose high service fees that discourage small-scale sellers and

bidders. Additionally, most existing systems do not offer seamless integration between product verification and bidding processes resulting in fragmented auction experiences.

Another major issue is the absence of transparency and security in bid handling, which can lead to fraudulent activities or unauthorized access to sensitive data. Smaller organizations and individual sellers also face challenges in finding affordable, easy-to-use auction tools that support secure user authentication in a single platform.

The BidMaster system aims to bridge these gaps by offering a unified, web-based solution that ensures real-time bidding updates, secure authentication and clear access for admins, sellers and bidders. By combining transparency and accessibility the platform provides a more efficient and trustworthy environment for conducting online auctions.

2.2 CHALLENGES IN EXISTING BIDDING TOOLS

Current online bidding systems face several limitations that affect their efficiency, usability, and reliability:

Complex Interfaces:

Many existing auction platforms have cluttered or outdated user interfaces that make navigation confusing for both sellers and bidders. This complexity often discourages user participation.

Lack of Real-Time Responsiveness:

Some systems fail to update bid information instantly, causing delays in reflecting the latest bid values. This not only disrupts the competitiveness of auctions but also reduces transparency and user trust in the bidding process.

Limited Security and Authentication:

Several platforms lack proper security measures such as two-factor authentication or encrypted data handling. As a result, they are vulnerable to fraudulent activities, unauthorized bidding and data breaches.

Poor Product Verification Process:

Many bidding tools do not verify the authenticity of listed items before they go live for auction. This can lead to the presence of fake or misleading product information, undermining the credibility of the platform.

Restricted Notification and Feedback Systems:

Most existing systems only provide basic email alerts or lack proper notification mechanisms for bid updates, auction start or end times, and result announcements, reducing user engagement and timely participation.

2.3 EFFICIENT BIDDING WITH A MODERN TOUCH

Our project aims to overcome the limitations of traditional and existing online auction systems by developing a modern, user-friendly, and secure bidding platform using Laravel 11 and Bootstrap 4. The system integrates multiple core features to provide a complete and efficient auction experience for both administrators and users.

- **Easy to Use:** Offers an intuitive and responsive interface that allows users to browse products, place bids, and view auction results effortlessly.
- **Comprehensive Auction Management:** Enables admins to create, manage, and monitor auctions efficiently, including product verification and bid tracking.
- **Real-Time Bidding:** Updates bid values instantly using dynamic web technologies, ensuring transparency and competitive interaction among bidders.
- **Secure Authentication:** Implements role-based access control and secure login systems to protect user data and maintain auction integrity.
- **Email Notification System:** Automatically notifies users about registration, auction outcomes, and important updates through integrated email services.

The BidMaster platform makes online bidding simpler, more accessible, and transparent. It provides a reliable environment for both sellers and bidders, combining real-time responsiveness, data security, and ease of use. By addressing the shortcomings of current systems, this project delivers a complete and efficient solution for modern online auctions.

3 OBJECTIVES

The core objectives of the Custom Auction and Product Platform are as follows:

1. To develop an online auction system where administrators can efficiently create auction listings.
2. To ensure secure user authentication and access control, distinguishing clearly between admin functions and bidder/Seller access.
3. To implement a real-time bidding mechanism that validates bids and updates the current bid price instantly for all participants.
4. To automatically determine and notify the winning bidder once an auction reaches its end time.
5. To maintain a simple and accessible user interface that encourages participation without unnecessary steps or complexities.

4 METHODOLOGY

4.1 REQUIREMENT IDENTIFICATION

4.1.1 LITERATURE REVIEW

Many existing online auction and bidding systems address only specific aspects of auction management, such as item listing or payment processing, without providing a fully integrated and secure environment for both sellers and buyers. Effective online auction platforms must manage multiple processes concurrently—user authentication, product verification, bid tracking, payment validation, and report generation—requiring coordination between system administrators, vendors and bidders. However, most existing systems lack seamless integration and adequate security mechanisms, often leading to data breaches, unauthorized bidding and inconsistent performance across platforms [1], [2], [3].

The emergence of eBay, founded in 1995 by Pierre Omidyar, marked the beginning of large-scale online auctioning. eBay evolved rapidly into a global e-commerce platform, hosting millions of listings and users worldwide. Despite its success, eBay faces challenges such as high transaction fees, rigid policies for sellers, and vulnerability to fraudulent listings and payment disputes [2], [4]. Similarly, eBid, established in 1998, offers an alternative model with lower selling fees and a user-friendly interface. While eBid provides improved cost efficiency and customer support, it attracts significantly less traffic, which affects the frequency and value of successful auctions [5].

In contrast, Listia, launched in 2009, introduced a credit-based exchange model where users trade goods using credits rather than real currency. This system promotes sustainability and user engagement but limits scalability due to its smaller user base and less intuitive credit conversion mechanism. The credit system also introduces complexity for users unfamiliar with non-monetary bidding [6], [7].

Previous academic studies have also explored web-based auction frameworks and their technical design. Aldaej et al. [1] developed a web-based auction system focusing on modular design and usability but noted shortcomings in authentication and data protection. Shirode et al. [8] proposed an online auction architecture emphasizing bidder verification, yet it lacked scalability for handling high-volume concurrent users. Similarly, Wen and Wong [9] examined bidder and seller behaviors under different auction models—ascending, descending, and Dutch auctions—highlighting that bidding psychology significantly affects transaction outcomes and system efficiency.

Historically, auction systems have evolved from manual, in-person bidding to fully automated web-based environments. Ancient civilizations like Babylon and Rome conducted physical auctions for trade and debt settlement [5]. Modern auction platforms, however, leverage digital communication technologies to enable global participation, real-time updates, and automated settlements [10]. Despite this evolution, the major challenges remain—ensuring secure transactions, validating bidder authenticity, and maintaining transparency in bid processing [3], [8].

To address these issues, the proposed BidMaster in this study integrates secure user registration and authentication for users(sellers and bidders) and admins. It introduces verification before listing products in auction and real-time bidding. Moreover, the system

aims to enhance scalability, data security and user experience through database-driven integration and web-based architecture. This approach not only improves on the limitations of existing systems but also promotes fair competition and ensures smoother auction management across all participants [10], [11].

4.1.2 REQUIREMENT ANALYSIS

To be able to run and develop the system or project, the minimum requirements for the system are:

- Operating system: Windows / macOS
- Internet browser: Google Chrome
- Web Server : Apache
- Backend Database : MySQL.
- Server Stack : XAMPP.
- Framework : Laravel 11.x
- Frontend : Bootstrap 4, Blade Template.
- PHPPackage Manager: Composer
- IDE/Text Editor : Visual Studio Code (VsCode)
- Email Service : SMTP server configuration (for sending email notifications)

4.2 DESIGN DIAGRAM

4.2.1 UML CLASS DIAGRAM

The class diagram describes the static structure of the bidding system, showcasing the system's various entities and their relationships. The key entities include AdminPanel, UserRecord, DocumentSubmission, Product, AuctionRecord, Bids.

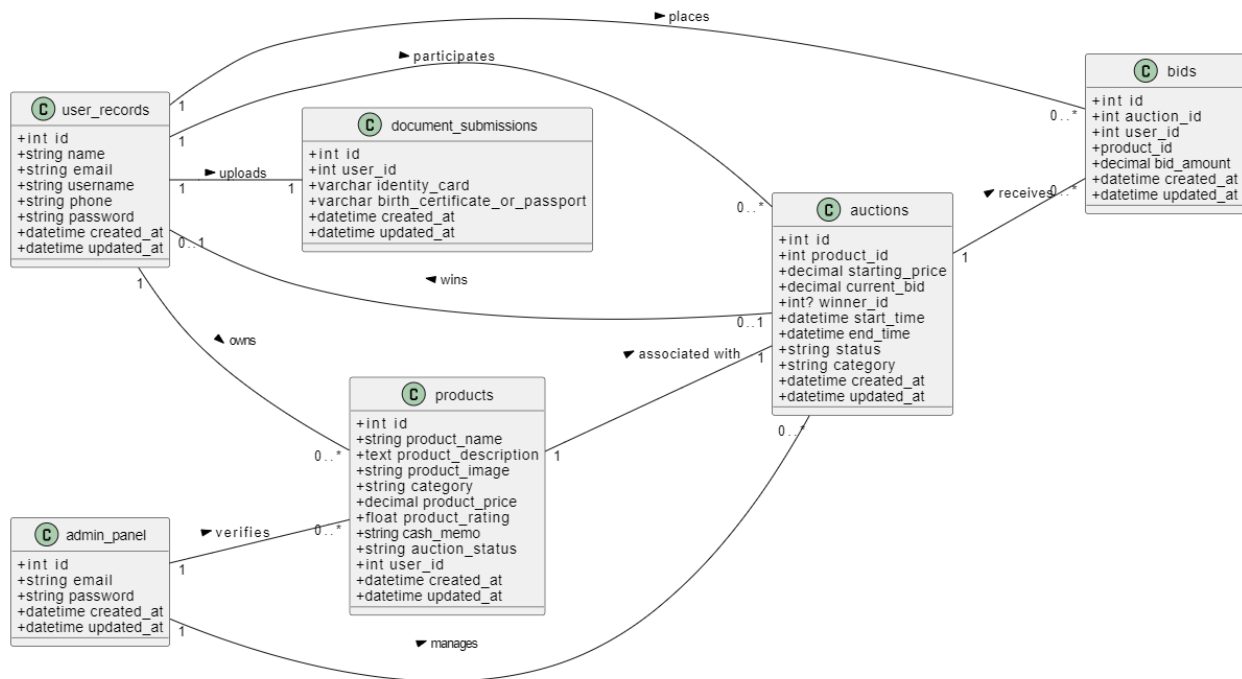


Figure 1: UML Class Diagram.

The class diagram clearly demonstrates in the bidding system how different entities interact, making it easier to understand the data flow and relationships within the system.

4.2.2 USE CASE DIAGRAM

The Use Case Diagram outlines all possible actions that Users and Admins can perform within the system. It visually represents the available features and interactions that users can have with the bidding platform.

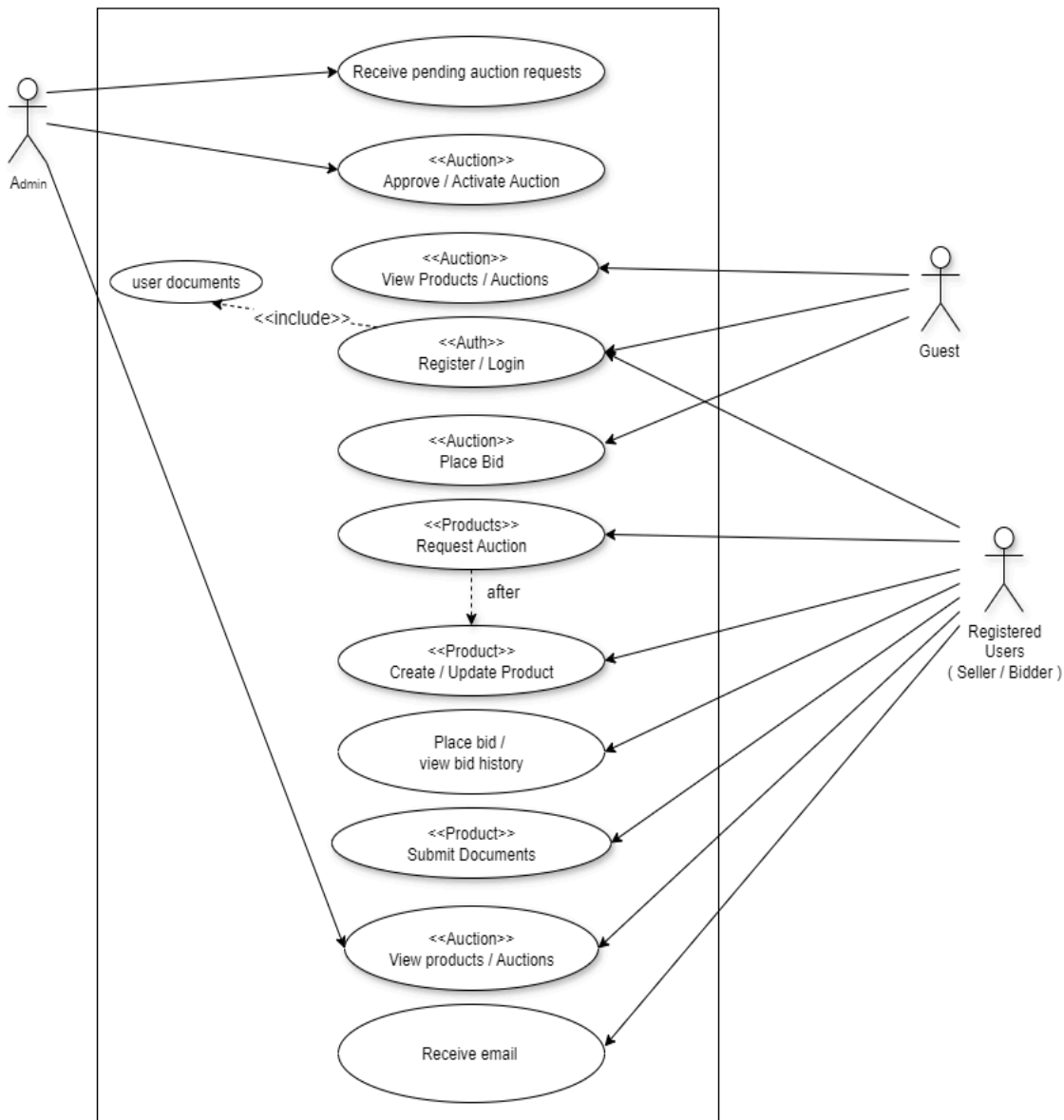


Figure 2: Use Case Diagram.

4.2.3 ENTITY-RELATIONSHIP DIAGRAM

The auction platform's data model is centered around these main entities—that work together to manage the entire process. The User_records entity represents all participants, including bidders and admins, with attributes like ID, name, email, and role. The Product entity defines each item listed for auction and includes an approval flag that must be true before an auction starts. The Auction entity tracks each auction's details, such as timing, current bid, and last bidder, while the Bid entity records every offer made by users, ensuring real-time updates and data consistency.

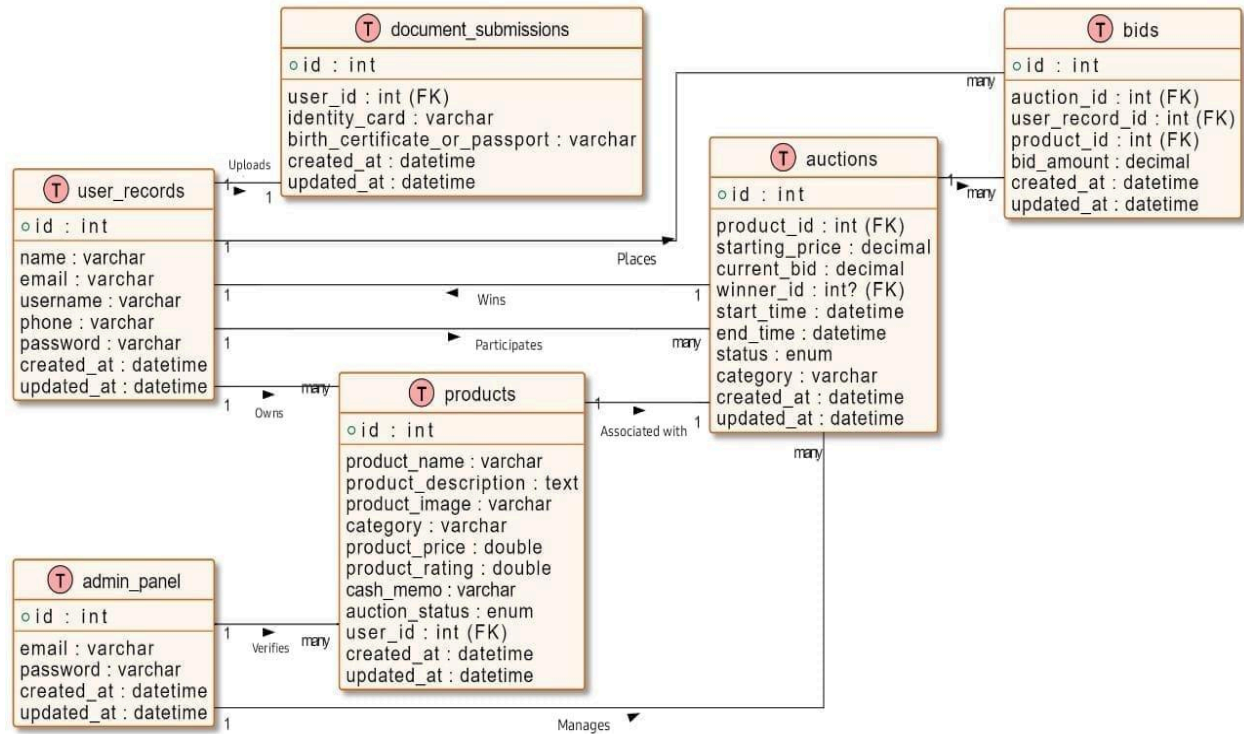


Figure 3: ER Diagram.

Key relationships maintain system integrity: A User can place many Bids, a Product can have multiple Auctions, and each Auction can contain many Bids. Together, these relationships form the foundation for secure, real-time auction operations.

4.2.4 ACTIVITY DIAGRAM

The activity diagram illustrates the overall workflow of the system, showcasing the interactions between users, admins, and products, from user registration and product submission to auction approval and active bidding, clearly distinguishing user activities from administrative operations.

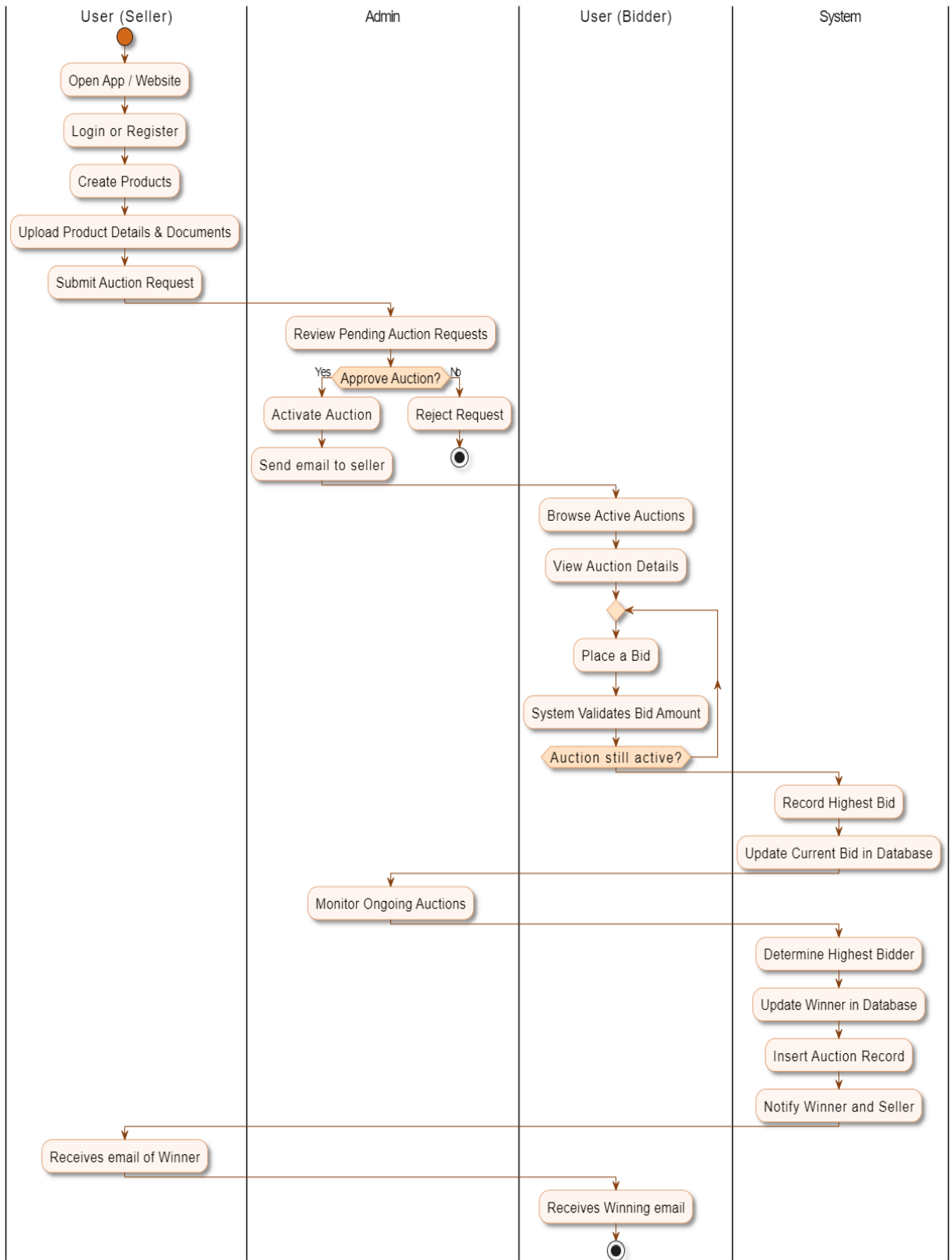


Figure 4: Activity Diagram.

For Admins:

- Admins sign in and access the Auction Management Panel and approve/reject products on the basis of product authenticity.
- After product approval, new auctions are created by specifying start time and end time.
- Once an auction is started, it remains active until the configured end time is reached.
- Admins' responsibility is creating and managing auctions.

For Bidders (Regular Users):

- When an auction is started, it automatically becomes visible and active on the Auction pages.
- Users can enter any active auction to view the current bid in real time and place their bids.
- The system validates each bid to ensure it is higher than the existing current bid. If valid, the system updates the bid amount instantly, making it visible to all participants.

For Sellers (Regular Users):

- Users (Seller) register and log in to the platform to list their products for auction. The seller submits product information and adds the product to auction.
- Once submitted, the product request is reviewed by the admin for authenticity and approval. Upon approval, the product becomes eligible for inclusion in an auction.
- During the auction, sellers can monitor real-time bidding activity on their listed items and view current bid prices and bid history. After the auction concludes, the seller is notified of the winning bid and bidder information through the system's email service, allowing them to proceed with further procedure.
- This bidding process continues until the auction's end time, at which point the highest bidder is declared the winner and notified via email.

4.2.5 SEQUENCE DIAGRAM

The Sequence Diagram explains the interaction flow between the entities and database. Below it covers how the users interact with the system.

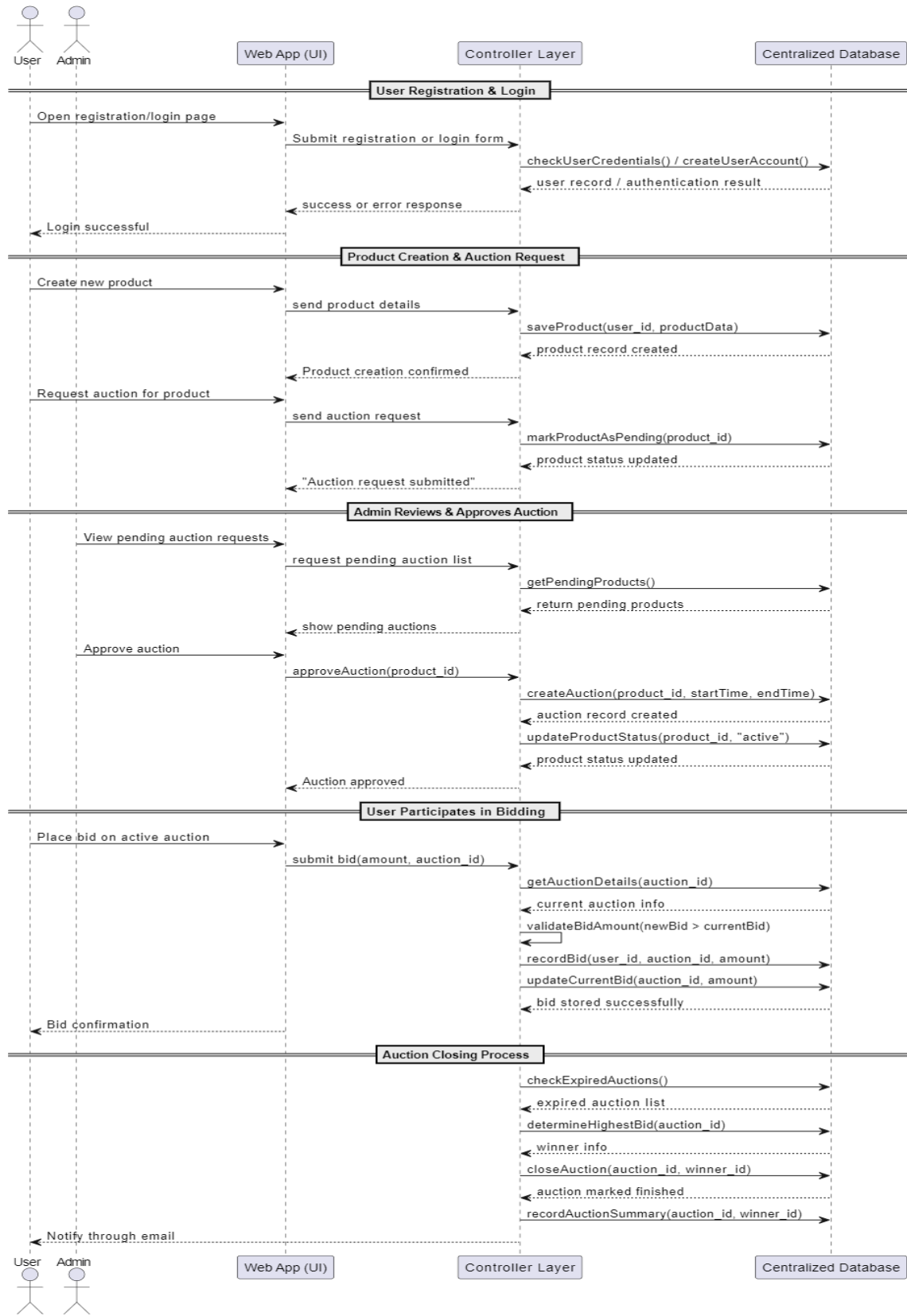


Figure 5: Sequence Diagram

4.2.6 DATA FLOW DIAGRAM

This DFD focuses on the movement of data between users, the database, and system components. It shows how information flows between different actors (admin and user) and the database that stores all critical information.

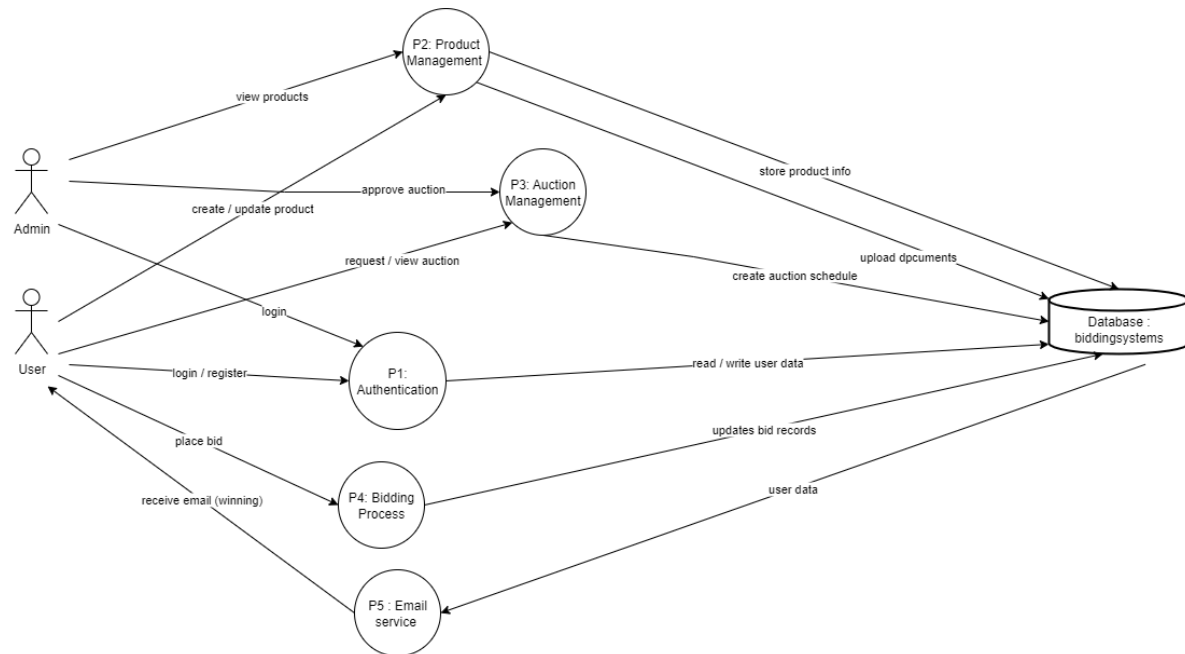


Figure 6: Data Flow Diagram.

The data flow diagram efficiently portrays the dynamic data flow within the system, showing how different user actions affect the data stored in the system's database. It highlights the system's reliance on data transactions for smooth operation.

4.3 FEASIBILITY STUDY

We have developed a system to manage bidding details and operations. Below is an overview of its estimated feasibility:

4.3.1 TECHNICAL FEASIBILITY

The required technology for the bidding system is easily available and compatible with the existing infrastructure. It can be integrated smoothly into current systems with minimal technical challenges.

4.3.2 OPERATIONAL FEASIBILITY

While some staff training will be necessary, it is straightforward and manageable. With proper planning, the system can be effectively integrated into existing operational workflows without major disruptions.

4.3.3 ECONOMIC FEASIBILITY

The economic feasibility analysis for the project evaluates the costs associated with the implementation and the potential benefits. Below is a detailed breakdown of the costs and benefits, followed by a payback period analysis.

Direct Costs

- **Capital Costs:**

- **Hardware:** TK 30,000 (for development machines and infrastructure)
- **Software Development and Licensing:** TK 38,000 (for development and licenses)
- **Training and Onboarding:** TK 10,000 (for user training)
- **Total Capital Cost:** TK 78,000

- **Recurring Costs:**

- **Maintenance and Support:** TK 2,000 per month Indirect Costs

- **Implementation Time:** Time and effort required for system transition.

- **Disruption:** Possible short-term disruptions during implementation.

Benefits

- **Tangible Benefits:**

- **Efficiency Gains:** Estimated monthly savings of TK 15,000 from improved efficiency.
- **Cost Reduction:** Additional savings of TK 20,000 per month through reduced operational costs.

- **Intangible Benefits:**

- **Improved Accuracy:** Enhanced data reliability and accuracy.
- **Enhanced User Experience:** Better user interface and increased satisfaction.

Payback Period Analysis:

- **Monthly Savings:** $TK (15,000 + 20,000) = 35,000$ (from efficiency and cost reductions).
- **Net Savings After Recurring Costs:** $TK 35,000 - TK 2,000 = TK 33,000$ per month.
- **Simple Payback:** $TK 78,000 / TK 33,000 = 2.36$ months(approx).

Payback with Interest:

- **Monthly Interest on Capital:** $1.2\% \times \text{TK } 78,000 = \text{TK } 936$
- **Net Savings After Interest:** $\text{TK } 33,000 - \text{TK } 936 = \text{TK } 32,064$
- **Adjusted Payback Period:** $\text{TK } 1,00,000 / \text{TK } 32,064 = 2.43 \text{ months.}$

Cost-Benefit Analysis :

•Present Value Method

Considering an interest rate of 1.2%

Assuming an interest rate of 1.2% per month, the present value of benefits acquiring n months later is calculated using the formula:

$$\text{Present Value} = x / (1 + r)^n$$

where: x is the benefit,

r is the interest rate and

n is the number of months.

Results:

Month	Cost (TK)	Net Benefit (TK)	Present Value (TK)	Cumulative Benefits (TK)
0	78000	0	0	0
1		32064	31671	31671
2		32064	31295	62966
3		32064	30924	93890
4		32064	30557	124447
5		32064	30195	154642

The analysis reveals that the cumulative benefits of the Bidding System exceed the costs in under 3 months. This Cost-Benefit Analysis demonstrates that investing in the platform is financially sound, with a swift payback period, ensuring cost recovery in a relatively short time frame. Both tangible and intangible benefits contribute significantly to:

1. Enhanced operational efficiency.
2. Improved user satisfaction.
3. Streamlined Auction processes.
4. Higher return on investment.

4.4 HIGHER LEVEL DESIGN OF SYSTEM

4.4.1 WORKING MECHANISM OF PROPOSED SYSTEM

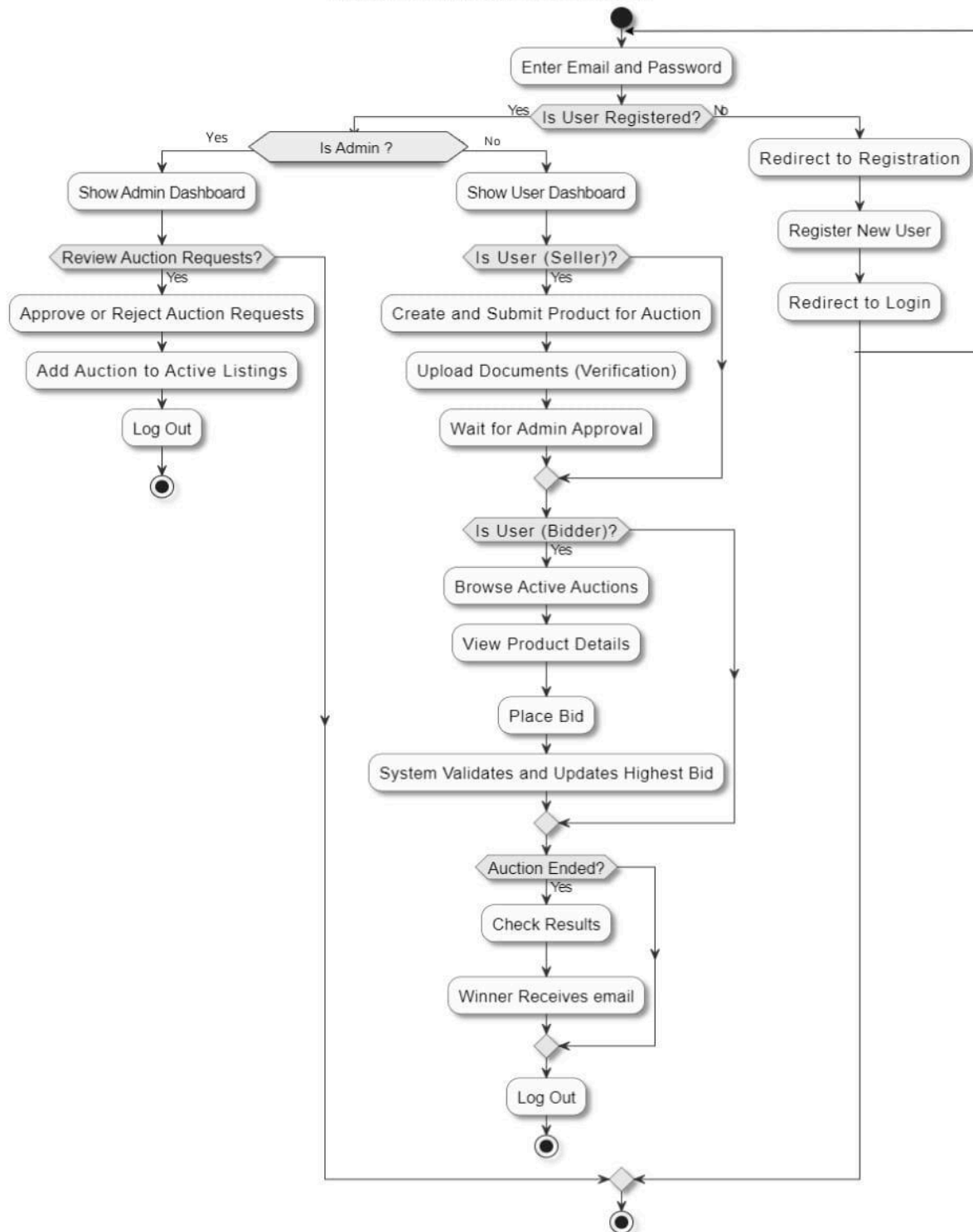
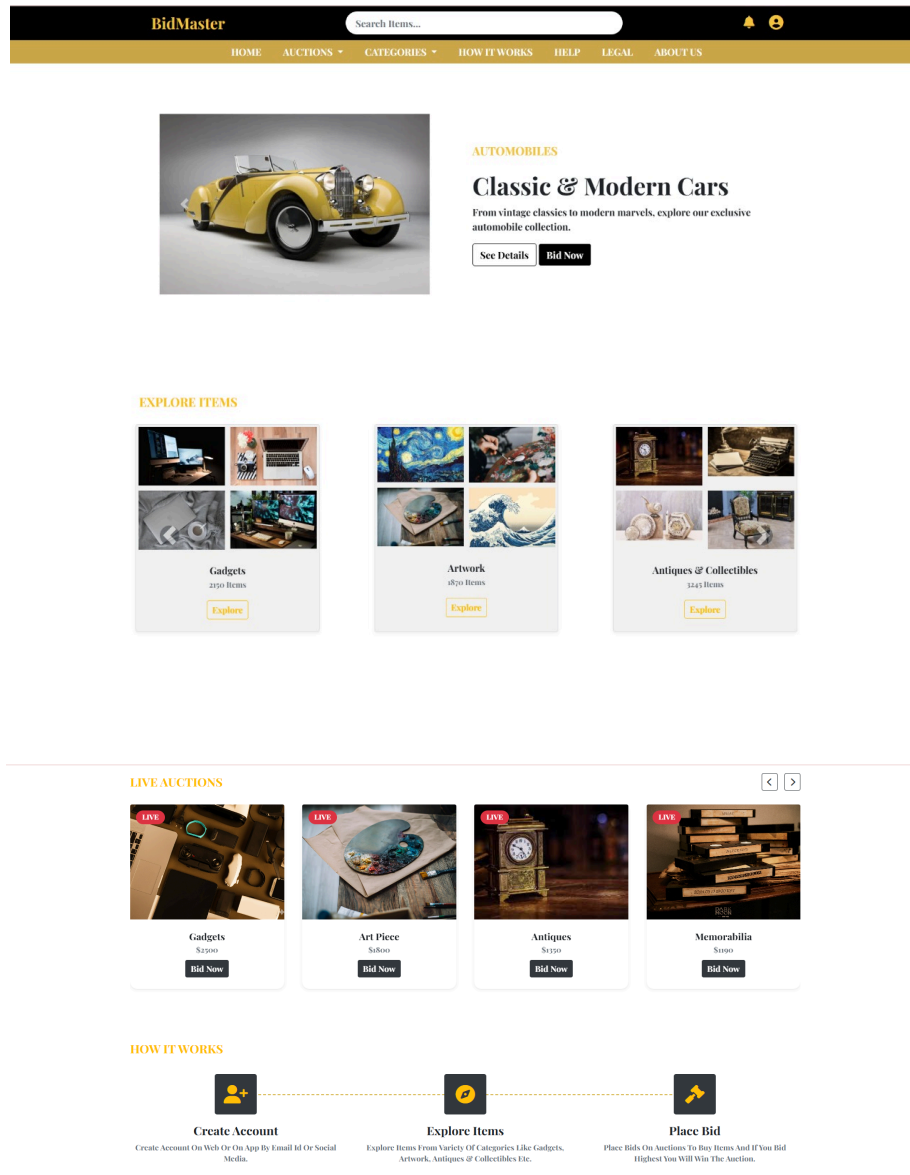
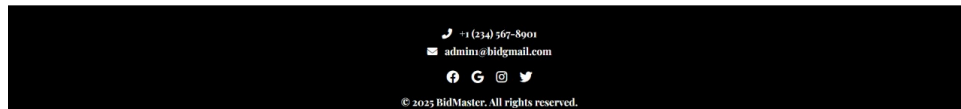
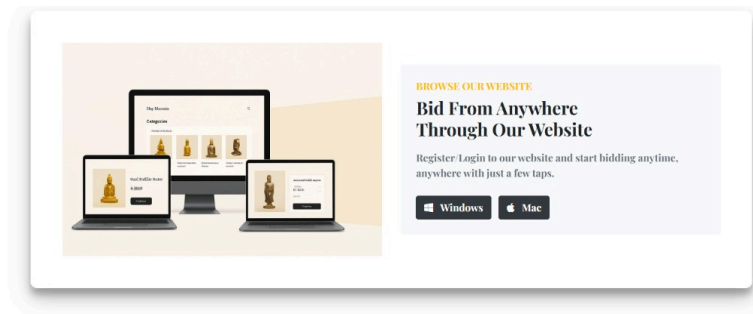


Figure 7: Higher Level Diagram.

5 IMPLEMENTATION

Home Page





This is the page users see when they visit our website. It gives an overview of our auction platform, highlights featured products and provides quick links to log in, register or explore items up for auction.

User Login/Register

Here, users can create a new account or log in to an existing one. Registration lets users participate in auctions, manage products and access their dashboard.

User Dashboard

The screenshot shows the BidMaster user dashboard. At the top, there's a navigation bar with the BidMaster logo, a search bar, and links for HOME, AUCTIONS, CATEGORIES, HOW IT WORKS, HELP, LEGAL, and ABOUT US. Below this, a grey bar displays the user's name 'MWalid' and email 'confidentialwalid@gmail.com' with a Logout button. The main content area is titled 'Your Listed Products' and contains a table with the following data:

Product	Category	Price	Auction Status
Signed Jersey	memorabilia	\$340	Approved
HP Pink Laptop	gadgets	\$400	Approved
ASUS Zenbook	gadgets	\$3400	None

At the bottom of the dashboard, a black bar displays a phone icon and the number '+1 (234) 567-8901'.

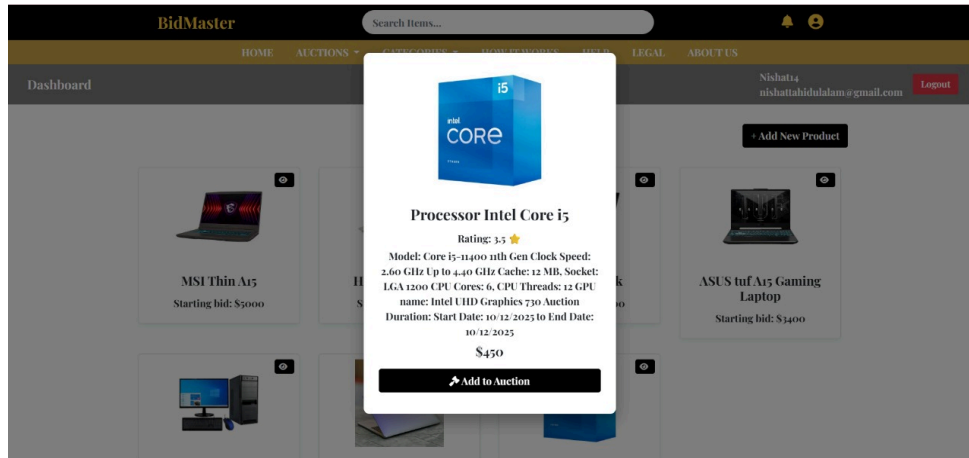
Once logged in, users land on their dashboard. It shows their active bids, won auctions, product listings, and bidding history. From here, they can navigate to add products or join live auctions.

Products

The screenshot shows the BidMaster products page. The navigation bar is identical to the dashboard. The main content area features a '+ Add New Product' button and four product listings, each with a car image, name, and starting bid:

- Mercedes-Benz**
Starting bid: \$45000
- Mercedes-Benz GLE**
Starting bid: \$56000
- Mercedes-Benz AMG**
Starting bid: \$55000
- Bugatti**
Starting bid: \$45000

Each listing includes a small 'Add' button with a plus icon and a 'Remove' button with a minus icon.



This section displays all products currently listed for auction. Registered users can add products to auction across five categories, browse through available items, check details like price and description, and choose which ones they want to bid on when the product goes onto auction.

Add Products

Add New Product

Product Name

Category

-- Select a Category --

Price (\$)

Rating (0-5)

Description (Please include the duration you want the auction to run*)

(Eg: Auction Duration: Start Date: MM/DD/YYYY to End Date: MM/DD/YYYY)

Product Image

Choose File No file chosen

Cash Memo

Choose File No file chosen

Add Product

Cancel

This page allows users to add new items to our auction platform. They can enter details like product name, description, images, and the starting bid amount.

Edit Products

Edit Product

Product Name

Mercedes-Benz GLE

Category

Automobiles

Price (\$)

560000

Rating (0-5)

4

Description (Please include the duration you want the auction to run*
(Eg: Auction Duration: Start Date: MM/DD/YYYY to End Date: MM/DD/YYYY)

Product Image

Choose File

No file chosen

Product Image

Cash Memo

Choose File

No file chosen

Existing Cash Memo:

Update Product

Cancel

Here, users can make changes to their existing product listings, such as updating descriptions, changing prices, or editing images before an auction if needed.

Admin Login

BidMaster

Search Items...

🔔

👤

HOME

AUCTIONS

CATEGORIES

HOW IT WORKS

HELP

LEGAL

ABOUT US

Admin Sign In

f

G+

in

or use your admin email for sign in

Email Address

Password

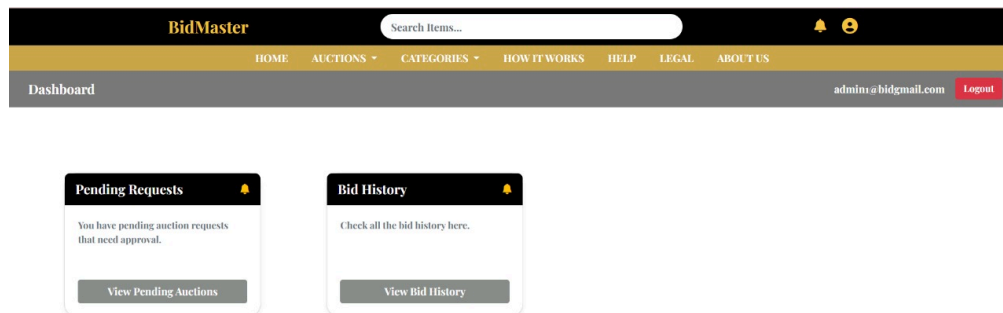
Log in to Dashboard

Admin Access Only

Please enter your admin credentials to manage the system

This is where our administrators sign in to access the backend of the website. It's a secure area meant only for authorized admin users.

Admin Dashboard



The admin dashboard gives admin full control over the platform. From here, admin can manage users, approve new products, monitor auctions, and keep track of system activities.

Pending Request

The screenshot shows the 'Pending Auction Requests' table in the Admin Dashboard. The table has the following data:

Product	User ID	Category	Price	Product PDF	User Documents	Action
Mercedes-Benz	1	automobiles	\$45000	View PDF	NID Birth Cert / Passport	Approve Reject
Processor Intel Core i5	5	gadgets	\$450	View PDF	NID Birth Cert / Passport	Approve Reject
Arch to Paradise	5	artwork	\$500	View PDF	NID Birth Cert / Passport	Approve Reject
Olympic Medals	5	memorabilia	\$32000	View PDF	NID Birth Cert / Passport	Approve Reject
Jersey	5	memorabilia	\$350	View PDF	NID Birth Cert / Passport	Approve Reject

This page lists all requests waiting for admin action for product approvals for an auction. Admins can approve or reject products from here.

Bid History (Admin Interface)

BidMaster

Search Items...

HOME

AUCTIONS

CATEGORIES

HOW IT WORKS

HELP

LEGAL

ABOUT US

Dashboard

admin@bidgmail.com

Logout

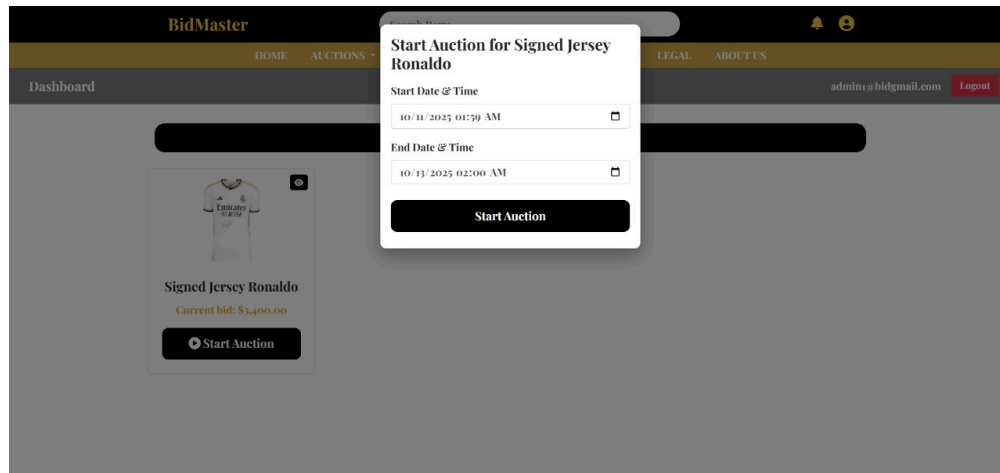
All Bid History

#	Product Name	Bidder	Bid Amount	Auction ID	Product ID	User Record ID	Bid Time
1	MSI Thin A15	MWalid	\$5,050.00	1	1	2	Oct 08, 2025 12:11 AM
2	Signed Jersey	Nishat	\$500.00	2	2	1	Oct 08, 2025 12:17 AM
3	MSI Thin A15	Nibedita	\$5,100.00	1	1	6	Oct 08, 2025 12:29 AM
4	Binoculars	MWalid	\$550.00	3	13	2	Oct 10, 2025 02:48 PM
5	Binoculars	Walid	\$560.00	3	13	3	Oct 10, 2025 03:48 PM

This section lets the admin review all bids placed by users. It helps admin track bidding activity, ensuring fairness, and maintaining transparency.

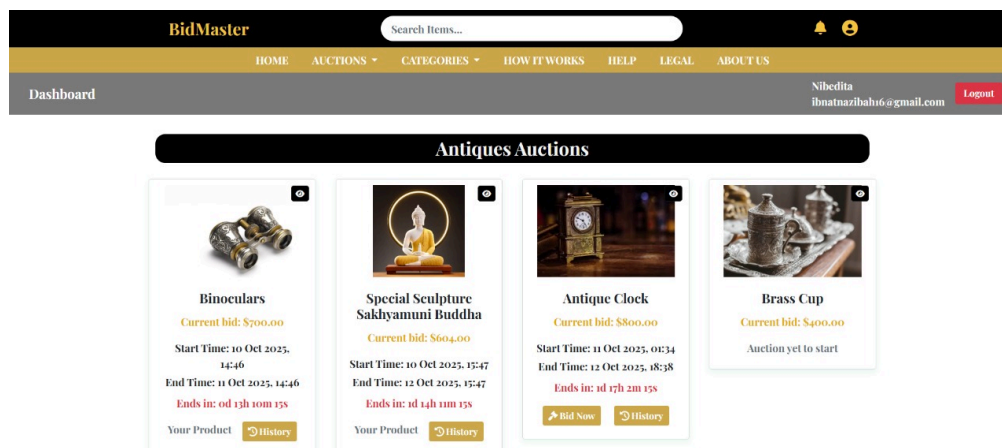
Auction Page (Admin Interface)

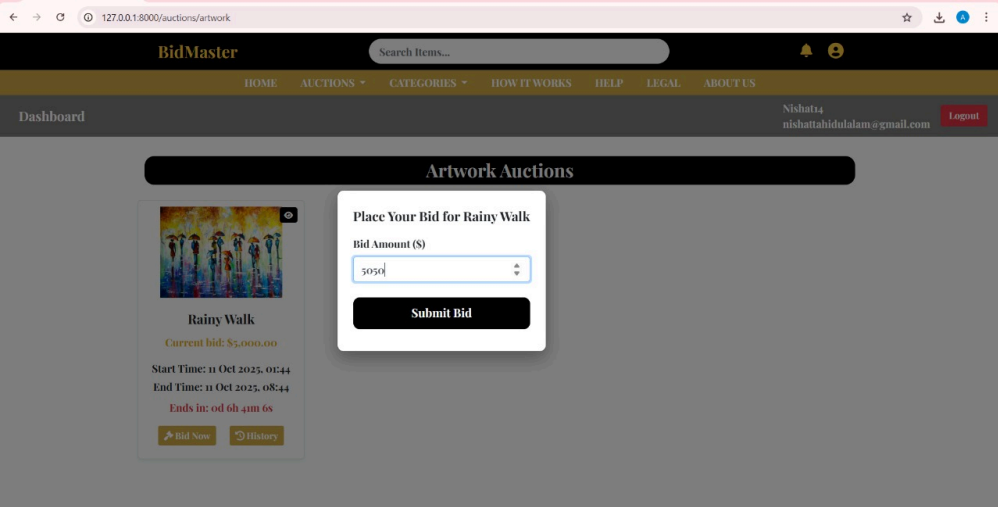
BidMaster		
Search Items...		
HOME AUCTIONS CATEGORIES HOW IT WORKS HELP LEGAL ABOUT US		
Dashboard admin@bidgmall.com Logout		
Antiques Auctions		
 Binoculars Current bid: \$700.00 Start Time: 10 Oct 2025, 14:46 End Time: 11 Oct 2025, 14:46 Ends in: 0d 13h 14m 36s	 Special Sculpture Sakhyamuni Buddha Current bid: \$604.00 Start Time: 10 Oct 2025, 15:47 End Time: 12 Oct 2025, 15:47 Ends in: 1d 14h 15m 36s	 Antique Clock Current bid: \$700.00 Start Auction



This is where the admin starts and manages live auctions. From here, the admin can set the auction start time, control its duration, monitor ongoing bids, and ensure that the auction runs smoothly until it ends.

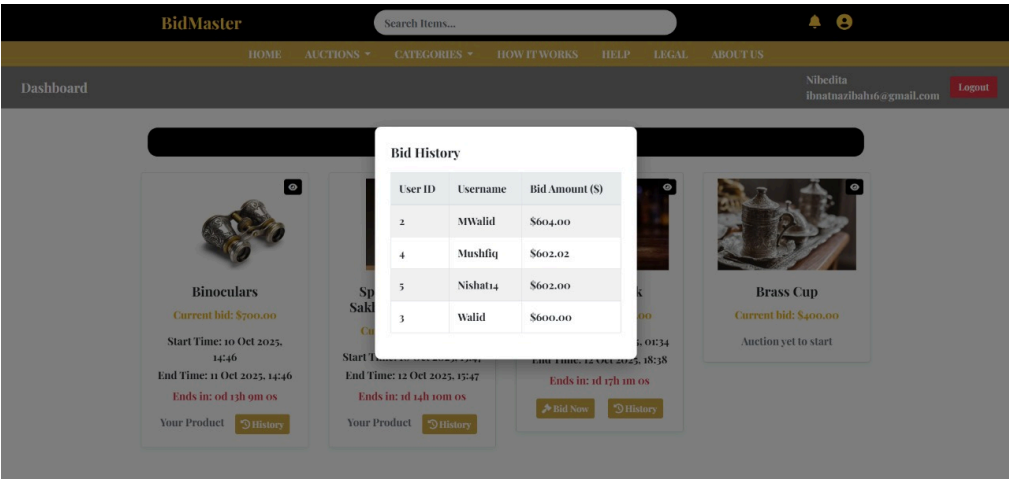
Auction Page (User Interface)





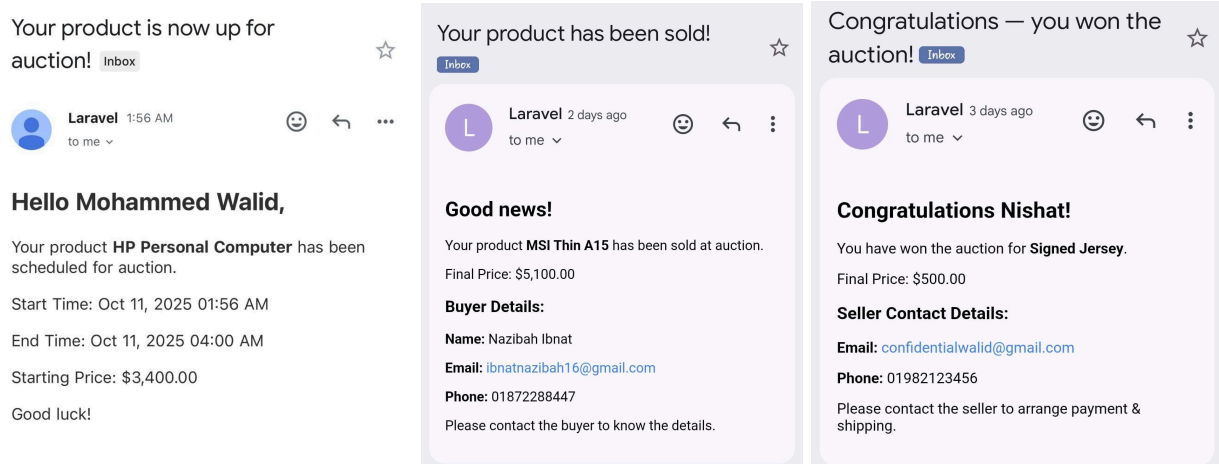
This is where users actively participate in live auctions. They can place bids, see real-time updates and track current highest bids on products they are interested in.

Auction Bidding History



Here, the number of bids placed on a specific product are displayed, along with a list of all the users who participated in bidding for that item.

Email



Our email system sends notifications for three auction events: when an auction starts, an email is sent to the product owner, when a product is sold, emails are sent to both the winning bidder and the product owner.

6 CONCLUSION

The detailed documentation of the BidMaster bidding system including different diagrams, the administrative activity flow, and supporting code structure—demonstrates a well-designed and reliable system architecture. It effectively incorporates key features such as admin auction approval to ensure marketplace security and real-time bidding through the different events for fast, responsive interactions. With the full blueprint now finalized, the project is ready for smooth implementation and testing of its core functionalities.

6.1 LACKINGS OF THE PROJECT

Although the BidMaster platform successfully meets its primary objectives of enabling secure, real-time online auctions, certain limitations remain that can be addressed in future iterations. These limitations mainly stem from resource constraints, time restrictions and the scope of the initial development phase. The key shortcomings are as follows:

Limited Payment Integration

The current version of the system does not include an integrated online payment gateway. All transactions are assumed to occur outside the platform, which limits the convenience and automation of post-auction settlements.

Absence of Mobile Application Support

The platform has been designed primarily for web browsers and does not currently provide a dedicated mobile application. As a result, accessibility and user engagement on mobile devices are not fully optimized.

Basic Notification System

Although email notifications are implemented for winner announcements, the system lacks advanced notification options such as real-time push notifications or SMS alerts for bid updates, auction reminders, and status changes.

Simplified User Interface for Admin Controls

While the admin panel includes all essential features, it lacks advanced analytics and reporting tools (e.g., auction performance graphs, user activity summaries) that could enhance decision-making and operational insights.

Limited Security Enhancements

Although Laravel provides a secure framework, additional layers such as two-factor authentication (2FA), encrypted bid data storage, and advanced intrusion detection mechanisms are not yet implemented.

6.2 FUTURE WORK

To enhance the functionality, scalability, and user experience of the BidMaster platform, several improvements can be considered for future development. These enhancements aim to address the current limitations and introduce advanced features that would make the system more robust, secure, and user-centric.

Integration of Secure Payment Gateway

Future versions of the system should include support for secure and automated online payment gateways such as Stripe, PayPal, or SSLCommerz. This integration would enable instant payment processing and verification after auctions, ensuring a seamless end-to-end transaction flow between bidders and sellers.

Development of a Mobile Application

To improve accessibility and user engagement, a dedicated mobile application can be developed for Android and iOS platforms. This would allow users to participate in auctions, receive notifications, and manage their accounts conveniently from any device, expanding the platform's reach and usability.

Enhanced Real-Time Notification System

The notification mechanism can be upgraded to include real-time push notifications, SMS alerts, and in-app notifications for critical events such as new bids, auction start or end times, and winning confirmations. Integrating these features with real-time web sockets or Firebase Cloud Messaging (FCM) would improve responsiveness and user interaction.

Advanced Admin Dashboard and Analytics

Future development should focus on extending the admin control panel to include data visualization and reporting modules. Features like graphical analytics of auction

performance, user participation trends, and revenue summaries would provide deeper insights for administrators and facilitate more informed decision-making.

Automated email notification

The system will include an automated email notification feature that sends a confirmation message to users immediately after successful registration. This addition will improve communication, verify user authenticity, and enhance the overall onboarding experience for new participants on the platform.

Strengthened Security Framework

To further enhance system security, future implementations could incorporate two-factor authentication (2FA), data encryption for bid and user information, and automated intrusion detection systems. Regular security audits and vulnerability testing would ensure data integrity and protect the platform from potential cyber threats.

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8 CONTRIBUTION

Mushfiqur Rahman

- Created How It Works and Help pages.

- Developed User Login / Logout functionality (backend).
- Built the User Dashboard.
- Designed and implemented the Footer section.

Nishat Afroz

- Designed the Home Page layout.
- Implemented User Middleware for authentication and access control.
- Designed and Developed the User Dashboard Navbar.
- Designed and Implemented Product CRUD operations (index, create, edit, delete).
- Created and designed Register/Login Page for users.
- Designed Product PDF in Pending Auction Request and Bid History functionality in admin dashboard.
- Designed and integrated Auction Admin Operations like start auction, selecting auction duration and auction countdown.
- Configured Email Notifications for Auction Started events.
- Contributed to UI/UX Design for overall platform aesthetics and usability.

Mohammed Walid

- Implemented Admin Middleware for authentication and access control.
- Designed the Live Auctions and How it Works section of the Home Page.
- Designed and developed the Admin Dashboard Navbar.
- Integrated Document Upload during User Registration.
- Designed the Products Pages on the basis of different categories with Add to Auction functionality.
- Managed Pending Auction Request Actions with Approve/Reject functionality in admin dashboard.
- Designed and integrated Auction User Operations like user bidding, user bid history.
- Configured Email Notifications for Auction Winner and Product Sold status updates.

Nazibah Ibnat Nibedita

- Developed Legal and About Us pages.
- Implemented Admin Login/Logout.
- Created Admin Seeder for initializing admin data.
- Designed and integrated the Header section.
- Built the Generate Document (PDF) feature in Pending Auction Request in the admin dashboard.
- Developed a Dynamic Search Bar for efficient product lookup.